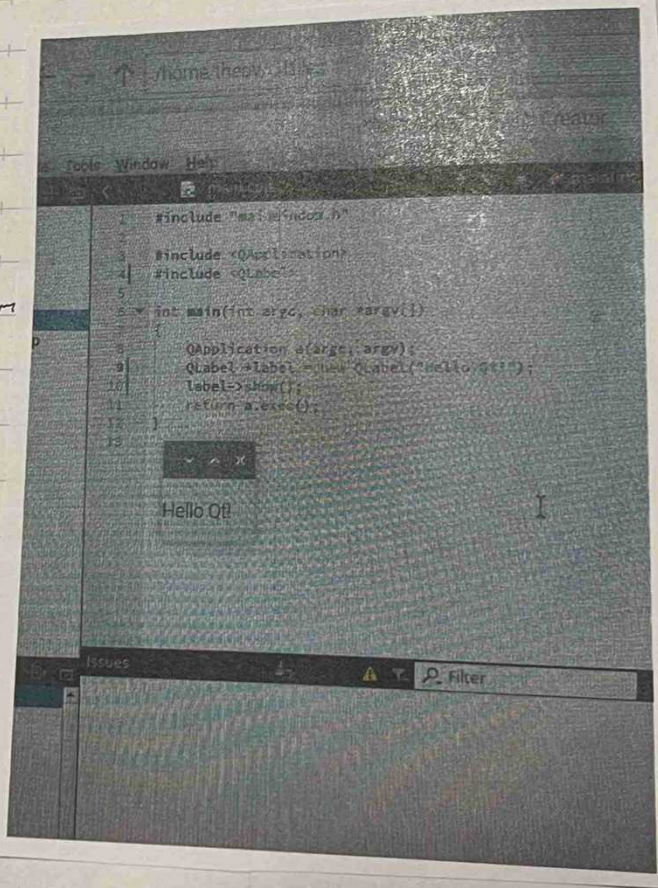


commonly used for developing user interfaces that divides the related program logic into three interconnected elements. These elements are the internal representations of information, the interface (the view) that presents information to and accepts it from the user, and the controller software linking the two.

source: <https://en.wikipedia.org/wiki/Model-view-controller>

2.3. Display a 'Hello Qt' message

- Copied and pasted the code shown in P6 laboratory document into the Qt Creator.
- Ran the code.



3. Laboratory Work

- Opened mainwindow.cpp.

3.1. Import the number

- Copied and pasted P6. zip into a newly created P6-3 inside P6 and unzipped it.

- Opened the P6-3 project in Qt Creator.

- Right Click on Source Filer → Add New... → C++ Class →

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update Class : tick all header files $\rightarrow$ Next $\rightarrow$ Ok
 - Added update.cpp and update.h to source files in CMakeLists.txt.

Lab Code:	None	Poor	OK	Good	Exceptional	Outstanding
Preparation	0	1	2	3	4	5
Progress	0	1	2	3	4	5
Understanding	0	1	2	3	4	5
Logbook Use	0	1	2	3	4	5
Student username:	fvlm23					5
Student Full name:	Timoteo Volante					Demonstrator only
						Signed: Bshaw

Went Post-laboratory Work

- Went to building 60 to continue the Qt project using RPi. 16/ March / 24
- It was unsuccessful : failed to read data from a text file.
- Went to my dormitory / halls of residence.
- Installed Qt on my laptop.
- Downloaded and unzipped the P6 sample code into the P6 folder.
- Opened Qt Creator and clicked new project.
- Named the new project as P6.proj inside the P6 folder.
- Copied and pasted the P6 sample code files into the P6.proj folder and replaced existing files.
- Created a text file named game and populated it with 9x9 1s.
- Made the code to read data from a text file in the Window constructor.
- Made the class Data to store the current state of the Sudoku game.
- Made the code to draw the current state of the Sudoku game on the screen in the paint Event() member function.

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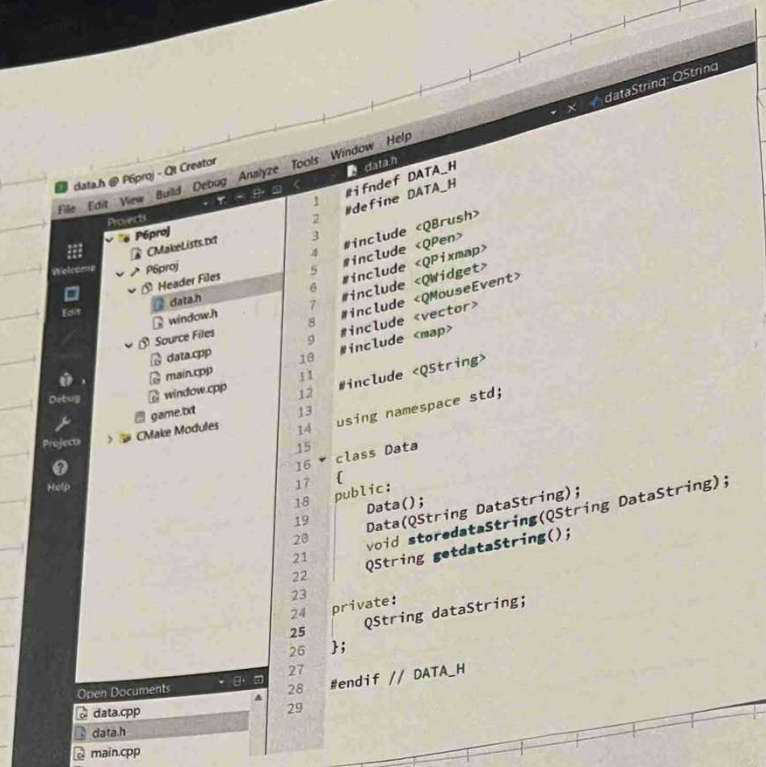
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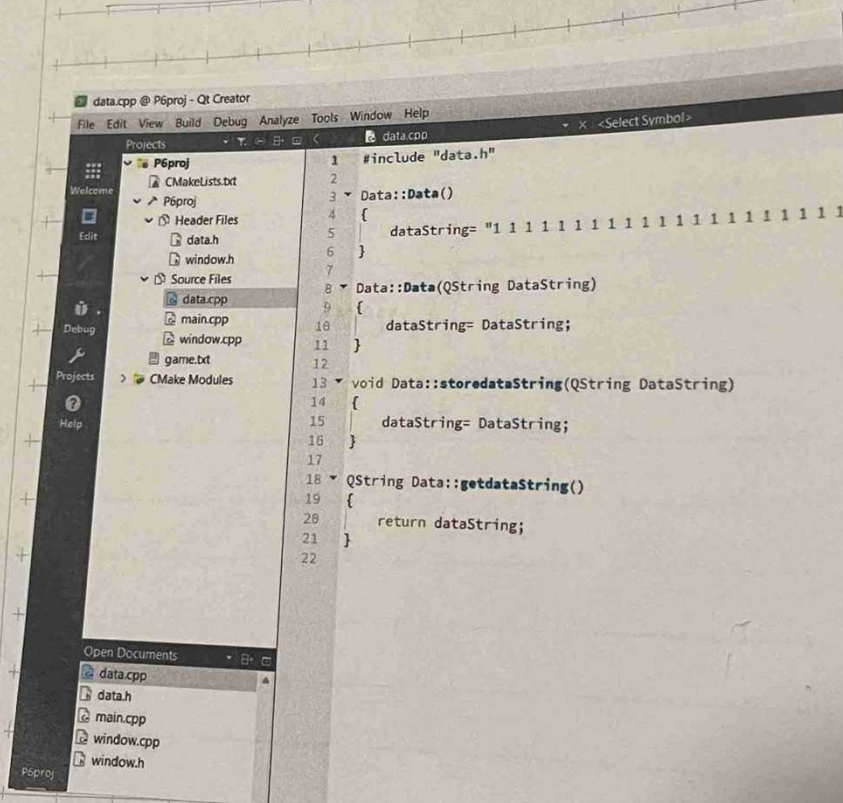
Date

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class Data
header file

```
class Data
store game n data.
```



class Data
cpp file

```

1 #include <iostream>
2 #include "window.h"
3 #include "data.h"
4
5 #include <QPainter>
6 #include <QDebug>
7 #include <QTextStream>
8 #include <QMessageBox>
9
10 #include <map>
11 #include <vector>
12
13 #include <QFile>
14
15 Data data1;
16
17 Window::Window(QWidget *parent) : QWidget(parent)
18 {
19     // set form size
20     setFixedSize(1000,1000);
21     setWindowTitle("Sudoku bla....");
22
23     QFile file("C:/Users/theov/OneDrive/Documents/EEE2623-2024/Semester2/Advanced Programming/Lab work/P6/P6proj/game.txt");
24     file.open(QIODevice::ReadOnly | QIODevice::Text);
25     QTextStream in(&file);
26     QString DataString;
27     DataStrings in.readAll();
28     data1.storedataString(DataString);
29     file.close();
30 }
31
32 void Window::paintEvent(QPaintEvent * event)
33 {
34     QPainter painter(this);

```

Widget Window
X
line 22 to 34
initializes the
object file for
the class Data
to store data
from it.

```

35
36
37
38 void Window::paintEvent(QPaintEvent * event)
39 {
40     QPainter painter(this);
41     QPen pen, pen2;
42     QFont font;
43     font.setPixelSize(30);
44     pen.setWidth(1);
45     pen2.setWidth(4);
46     painter.setPen(pen);
47     painter.setFont(font);
48     QPoint p1(10,20), p2(80,60);
49     painter.drawLine(p1, p2);
50     painter.setPen(pen2);
51     QPoint p3(80, 20), p4(80,900);
52     painter.drawLine(p3, p4);
53     QPoint p5(20, 60), p6(900, 60);
54     painter.drawLine(p5, p6);
55
56     QString DataString= data1.getdataString();
57     QStringList DataList= DataString.split(" ");
58     for(int i=0; i<81; i++)
59     {
60         for(int j=0; j<9; j++)
61         {
62             if((j==2)|(j==5)|(j==9)) { painter.setPen(pen2);}
63             else {painter.setPen(pen);}
64             QPoint p1(150+j*90, 20), p2(150+j*90,900);
65             painter.drawLine(p1, p2);
66             for(int k=0; k<9; k++)
67             {
68                 painter.setPen(pen2);
69                 painter.drawText(100 + j*90,100 + k*80, DataList[i]);
70             }
71         }
72     }
73     for(int k=0; k<9; k++)
74     {
75         if((k==2)|(k==5)|(k==9)) { painter.setPen(pen2);}
76         else {painter.setPen(pen);}
77         QPoint p1(20, 130+k*80 ), p2(900,130+k*80);

```

paintEvent()
method "painter"
the GUI
and updates
it when the
event is triggered.

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```

54 painter.drawLine(p5, p6);
55
56 QString DataString= data1.getDataString();
57 QStringList DataList= DataString.split(" ");
58 for(int i=0; i<81; i++)
59 {
60     for(int j=0; j<9; j++)
61     {
62         if((j==2)|(j==5)|(j==9)) { painter.setPen(pen2);}
63         else {painter.setPen(pen);}
64         QPoint p1(150+j*90, 20), p2(150+j*90,900);
65         painter.drawLine(p1, p2);
66         for(int k=0; k<9; k++)
67         {
68             painter.setPen(pen2);
69             painter.drawText(100 + j*90,100 + k*80, DataList[i]);
70         }
71     }
72 }
73 for(int k=0; k<9; k++)
74 {
75     if((k==2)|(k==5)|(k==9)) { painter.setPen(pen2);}
76     else {painter.setPen(pen);}
77     QPoint p1(20, 130+k*80 ), p2(900,130+k*80);
78     painter.drawLine(p1, p2);
79 }
80 painter.setPen(pen2);
81 QPoint p7(870, 20), p8(870, 820);
82 painter.drawLine(p7, p8);
83 QPoint p9(20, 770), p10(900, 770 );
84 painter.drawLine(p9, p10);
85 }
86
87 void Window::mouseReleaseEvent(QMouseEvent * event)
88 {
89     // get click position
90     qDebug() << "Mouse x " << event->x() << " Mouse y " << event->y();
91 }
92

```

Sudoku GUI

1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1

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```

168 And:: And() {a=0; b=0;}
169 And:: And(int A, int B) {a=A; b=B;}
170 int And::output() { res= a * b; return res;}
171 Or:: Or() {a=0; b=0;}
172 Or:: Or(int A, int B) {a=A; b=B;}
173 int Or::output() { res= a + b; return res;}
174 Xor:: Xor() {a=0; b=0;}
175 Xor:: Xor(int A, int B) {a=A; b=B;}
176 int Xor::output() { res= a ^ b; return res;}
177
178 halfAdder:: halfAdder(){a=0; b=0; sum= 0; carryout=0;}
179 halfAdder:: halfAdder(int A, int B, int Sum, int Carryout){
180     a= A; b= B; sum= Sum; carryout= Carryout;}
181 xor= Xor(a, b);
182 and= And(a, b);
183 sum = xor.output();
184 carryout= and.output();
185 int halfAdder:: getsum(){return sum;}
186 int halfAdder:: getcarryout(){return carryout;}
187 void halfAdder:: setsum(int Sum){sum= Sum;}
188 void halfAdder:: setcarryout(int carryout){carryout= Carryout;}
189
190 fullAdder:: fullAdder(){a=0; b=0; sum1= 0; carryin1=0; carryout1=0;}
191 fullAdder:: fullAdder(int A, int B, int Sum1, int Carryin1, int Carryout1){
192     a=A; b=B; sum1=Sum1; carryin1=Carryin1; carryout1=Carryout1;}
193 h1= halfAdder(a, b, 0, 0);
194 sum1= h1.getsum();
195 carryin1= h1.getcarryout();
196 h2= halfAdder(carryin1, sum1, 0, 0);
197 sum1= h2.getsum();
198 carryin2= h2.getcarryout();
199 or= Or(carryin1, carryin2);
200 carryout1= or.output();
201
202 fullAdder:: getsum1(){return sum1;}
203 fullAdder:: getcarryout1(){return carryout1;}
204 fullAdder:: getcarryin1(){return carryin1;}
205 fullAdder:: setsum1(int Sum1){sum1= Sum1;}
206 fullAdder:: setcarryout1(int Carryout1){carryout1= Carryout1;}
207 fullAdder:: setcarryin1(int Carryin1){carryin1= Carryin1;}

```

P6: Soda Sudoku GUI 1

2: Preparation

13/ March / 24

2.1. Prepare SD Card

- Inserted the card reader with the SD card into my laptop.
- Read the Raspberry Pi Qt Installation Guide.
- Downloaded and installed Pi Imager.
- Chose Raspberry Pi 4 Models B ... for the device, 'bookworm' Raspberry pi OS (64-bit) for the operating system, and the SD card for the storage, and clicked next.
- 'Written' the SD card.
- Removed the SD card from the laptop.
- Inserted the SD card into the RPi, and powered it up.
- Downloaded Raspberry Pi Desktop in my laptop.
- Downloaded balena Etcher.
- Read <https://projects.raspberrypi.org/en/projects/install-raspberry-pi-desktop/2> to install RPi Desktop to laptop.
- Clicked Flash from file and clicked the downloaded RPi Desktop file.

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- It failed because the USB Drive does not have enough storage.
- Rewritten the SD card with the card reader on my laptop, but this time, configured the settings.
- Host name: raspberry
- Username: Neo
- Configure wireless LAN SSID: eduroam and password
- Enabled SSH tick Use password authentication in OS customisation
- Clicked write SAVE and clicked Yes to apply OS customisation settings.
- Started the writing process.
- Finished the writing process.
- Verified the flashing.
- Inserted the SD card, plugged the laptop to the RPi₁ and powered the RPi.
- Downloaded putty.exe 64-bit x86 (the SSH and Telnet client (set) from putty.org.
- Downloaded Xming X Server from <https://sourceforge.net/projects/xming/>
- Ran Xming as administrator and, installed, and clicked finished.
- Turned the laptop off, turned it on, immediately pressed Esc and F9 to go to Boot menu.
- Watched videos using the headless method to set-up RPi and tried them but failed.
- Went to building 60 of the University of Southampton 14/ March /21 Highfield Campus.
- Attached the keyboard, mouse, hdmi cable between RPi and the computer.
- Attached Powered the RPi up.
- Booted the RPi.
- The RPi screen showed up.
- Clicked Advanced Options → Edit Connections → Wireless → wireless security → WPA / WPA 2 / WPA3 Personal on WiFi Menu.

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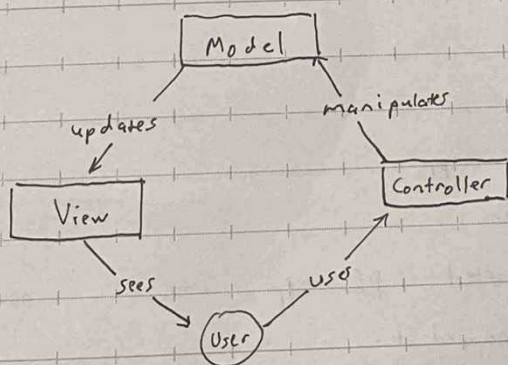
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- Clicked eduroam and completed the ^{connection} configuration as shown in the Raspberry Pi Qt Installation Guide and clicked Save.
- Successfully connected to eduroam.
- Downloaded the required files on RPi Desktop into a created folder, Qt files, and unzipped it.
- Typed and entered ls in the command line.
- " " " cd RPi Qt files
- " " " chmod +x run_bookworm.sh
- " " " ls
- " " " ./run_bookworm.sh
- Ran the installation script.
- Checked if Qt is on the menu.
- Opened Qt Creator.
- Clicked Create Project.
- Chose Qt Widgets Application.
- Named it P6 and created in /home/Heav.
- Build System: CMake.
- Verified classmate and clicked next.
- Clicked next for the rest of the prompts.
- Ran the default code by clicking the 'play' ▶ green button.
- A blank Main Window showed up.

2.2. Read about programming GUIs



source: <https://en>

source: Model-view-controller (MVC) is a software design software

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Post-laboratory Work

- Remade the code of the sudoku.
- Removed the class Data.

```

1 #include <QBrush>
2 #include <QPen>
3 #include <QPixmap>
4 #include <QWidget>
5 #include <QMouseEvent>
6 #include <QPushButton>
7 #include <QFileDialog>
8
9
10
11 #include <vector>
12 #include <map>
13
14 #include <QLabel>
15 #include <QVector>
16
17
18 using namespace std;
19
20 class Window : public QWidget
21 {
22     Q_OBJECT
23
24 public:
25     Window(QWidget *parent = 0);
26
27 protected:
28     // methods and members
29     void paintEvent(QPaintEvent *event);
30     QVector<QPushButton*> plsBtnList;
31     QVector<QPushButton*> minBtnList;
32     QString lbl;
33     QStringList lbl2;
34     QStringList data;
35
36 public slots:
37     // event handlers (in Qt "slots")
38     // ...
39     void Open();
40     void Plus();
41     void Minus();
42     void Check();
43 };

```

line 20-31

- Added plsBtn List, and minBtn List dynamic or a list of QPushButton.

line 32

- Added QString lbl to display check() output : win or lose incorrect

line 33

- Added QStringList lbl2 to mark "*" on initial non-zero cells : they can't be changed in the game.

line 34

- Added QStringList data to store number data that change when clicks up and down buttons are clicked.

line 29-36 :

initialization of data and lbl2

```

1 #include "window.h"
2
3 #include <QPainter>
4 #include <QDebug>
5 #include <QTextStream>
6 #include <QMessageBox>
7 #include <QGridLayout>
8 #include <QSpinBox>
9 #include <QPushButton>
10 #include <QFileDialog>
11
12 #include <map>
13 #include <vector>
14
15 #include <QFile>
16 #include <QVector>
17
18 using namespace std;
19
20
21 // Main window class
22 Window::Window(QWidget *parent) : QWidget(parent)
23 {
24     // set form size
25     setFixedSize(1000, 1000);
26     setWindowTitle("Bla...");
27
28     // create widgets....
29     for(int i=0; i<81; i++)
30     {
31         data.append("0");
32     }
33     for(int i=0; i<81; i++)
34     {
35         lbl2.append(" ");
36     }
37
38     // open button
39     auto *openBtn = new QPushButton("Open", this);
40     openBtn->setGeometry(900, 450, 75, 30);
41 }

```

ne _____ Date

ne _____ Date

```

40 connect(openBtn, &QPushButton::clicked, this, &Window::Open);
41
42 //plusminus buttons
43 int l=0;
44
45 for (int k=0; k<9; k++)
46 {
47     for (int j=0; j<9; j++)
48     {
49         if(data[l]=="")
50         {
51             auto *plsBtn= new QPushButton("+", this);
52             auto *minBtn= new QPushButton("-", this);
53             plsBtnList.append(plsBtn);
54             minBtnList.append(minBtn);
55
56             plsBtnList[l]->setGeometry(130 + j*90, 90 + k*80, 20, 20);
57             minBtnList[l]->setGeometry(130 + j*90, 110 + k*80, 20, 20);
58
59             plsBtnList[l]->setProperty("order", l);
60             minBtnList[l]->setProperty("order", l);
61
62             connect(plsBtnList[l], &QPushButton::clicked, this, &Window::Plus);
63             connect(minBtnList[l], &QPushButton::clicked, this, &Window::Minus);
64             l++;
65         }
66     }
67 }
68
69 //check button
70 auto *checkBtn= new QPushButton("Check", this);
71 checkBtn->setGeometry(900, 500, 75, 30);
72
73 connect(checkBtn, &QPushButton::clicked, this, &Window::Check);
74 }
75
76 void Window::Plus() {
77     QPushButton *plsBtn= qobject_cast<QPushButton*>(sender());
78     int ord= plsBtn->property("order").toInt();
79

```

line 44 - 74
~~Added~~ ~~Re Added~~
 Replaced
 initialization of
 dynamic array of
 up and down buttons

lines 76 - 96
~~Replaced~~ Modified
 plus() and minus()
 buttons for
 the new member,
 data, and set
 max and min as 9
 and 0.

```

79 int val= data[ord].toInt();
80 if(val!=9){val++;}
81
82 data[ord]= QString::number(val);
83 update();
84 }
85
86 void Window::Minus() {
87     QPushButton *plsBtn= qobject_cast<QPushButton*>(sender());
88     int ord= plsBtn->property("order").toInt();
89
90     int val= data[ord].toInt();
91     if(val!=0){val--;}
92
93     data[ord]= QString::number(val);
94     update();
95 }
96
97 void Window::paintEvent(QPaintEvent * event)
98 {
99     QPainter painter(this);
100     QPen pen, pen2;
101     QFont font;
102     font.setPixelSize(30);
103     pen.setWidth(1);
104     pen2.setWidth(4);
105     painter.setPen(pen);
106     painter.setFont(font);
107
108     //leftmost line and topmost line
109     painter.setPen(pen2);
110     QPoint p3(70, 20), p4(70, 900);
111     painter.drawLine(p3, p4);
112     QPoint p5(20, 60), p6(900, 60);
113     painter.drawLine(p5, p6);
114
115     //QStringList dataList= datal.getdata();
116     int i=0;
117     for (int k=0; k<9; k++)
118

```

```

window.cpp @ P7proj - Qt Creator
File Edit View Build Debug Analyze Tools Window Help
Projects
P7proj
  CMakeLists.txt
  P7proj
    Header Files
      window.h
    Source Files
      main.cpp
      window.cpp
      game.txt
      mainwindow.ui
  CMake Modules
  Open Documents
    main.cpp
    window.cpp
    window.h
  P7proj
  Release

119 for (int k=0; k<9; k++)
120 {
121     for (int j=0; j<9; j++)
122     {
123         if (k==0) {
124             if ((i==2) || (i==5) || (i==8)) { painter.setPen(pen2);
125                 else { painter.setPen(pen1);
126                 QPoint pl(150+j*20, 20), pr(150+j*20, 80);
127                 painter.drawLine(pl, pr);
128             }
129             painter.setPen(pen2);
130             painter.drawText(100 + j*20, 100 + k*20, data[i]);
131             painter.drawText(120 + j*20, 100 + k*20, lbl[i]);
132             i++;
133         }
134         if ((k==2) || (k==5) || (k==8)) { painter.setPen(pen2);
135             else { painter.setPen(pen1);
136             QPoint pl(20, 130+k*20), pr(980, 130+k*20);
137             painter.drawLine(pl, pr);
138             i++;
139         }
140     }
141     //rightmost line and bottommost line
142     painter.setPen(pen2);
143     QPoint pl(870, 20), pr(870, 820);
144     painter.drawLine(pl, pr);
145     QPoint pl(20, 770), pr(980, 770);
146     painter.drawLine(pl, pr);
147     font.setPointSize(20);
148     if (lbl == "You won!") { painter.drawText(880, 450, lbl);
149         else if (lbl == "You lost!") { painter.drawText(880, 450, lbl);
150     }
151 }
152 void Window::Open() {
153     QString filename = QFile::fileName().replace("P7proj", "P7proj");
154     if (!QFile::exists(filename)) {
155         tr("Open File", "C:/Users/Name/Desktop/Document/2024/Semester2/Advanced Programming/lab work/P7proj", tr("Text Files (*.txt)"));
156         QFile file(filename);
157     }
158 }

```

paint Event ()
line 129 to 130.

- Replace DataList[i]
with data[i]
and added lbl[i]
to display the "*" on non-zero cells

```

window.cpp @ P7proj - Qt Creator
File Edit View Build Debug Analyze Tools Window Help
Projects
P7proj
  CMakeLists.txt
  P7proj
    Header Files
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      main.cpp
      window.cpp
      game.txt
      mainwindow.ui
  CMake Modules
  Open Documents
    main.cpp
    window.cpp
    window.h
  P7proj
  Release

156 file.open(QIODevice::ReadOnly | QIODevice::Text);
157
158 QTextStream in(&file);
159
160 QString DataString;
161
162 DataString = in.readAll();
163
164 QStringList DataList = DataString.split(",");
165
166 data = DataList;
167
168 //disconnect signals with initial nonzero value and reconnects them with zero value
169 int i=0;
170 for (int k=0; k<9; k++)
171 {
172     for (int j=0; j<9; j++)
173     {
174         if (data[i]!="0")
175         {
176             disconnect(plsBtnList[i], &QPushButton::clicked, this, &Window::Plus);
177             disconnect(minBtnList[i], &QPushButton::clicked, this, &Window::Minus);
178             lbl2[i]="*";
179         }
180         else
181         {
182             if ((disconnect(plsBtnList[i], &QPushButton::clicked, this, &Window::Plus) &&
183                 disconnect(minBtnList[i], &QPushButton::clicked, this, &Window::Minus)) == true)
184             {
185                 connect(plsBtnList[i], &QPushButton::clicked, this, &Window::Plus);
186                 connect(minBtnList[i], &QPushButton::clicked, this, &Window::Minus);
187             }
188             lbl2[i]=" ";
189         }
190         i++;
191     }
192     i++;
193 }
194 lbl=" ";

```

Open()

line 167 data stores data from text file

line 170 - 194 Disconnects signals for non zero cells

and initialize lbl2[i] as "*". Reconnects signals for

zero cells once a new game is opened, and reinitialize lbl2[i]
to " " for zero cells.

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```

window.cpp @ P7proj - Qt Creator
File Edit View Build Debug Analyze Tools Window Help
Projects
P7proj
  CMakeLists.txt
  P7proj
    Header Files
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    Source Files
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      window.cpp
      game.txt
      mainwindow.ui
  CMake Modules
Open Documents
main.cpp
window.cpp
window.h
P7proj
Release

194     lbl=" ";
195     file.close();
196 }
197
198 void Window::Check() {
199
200     //check for horizontal sum
201     int sum=0;
202     int i=0;
203     int sg0=0, sg1=0, sg2=0;
204     for(int k=0; k<9; k++)
205     {
206         for(int j=0; j<9; j++)
207         {
208             sum += data[i].toInt();
209             i++;
210         }
211         if(sum!=45) { lbl= "Incorrect";sg0=1;
212             update(); }
213         else{ sum=0; }
214     }
215     // qDebug()<<sg0;
216     // check for vertical sum
217     sum=0;
218     for(int j=0; j<9; j++)
219     {
220         for(int k=0; k<9; k++)
221         {
222             sum += data[j+(9*k)].toInt();
223         }
224         if(sum!=45) { lbl= "Incorrect";sg1=1;
225             update(); }
226         else{ sum=0; }
227     }
228
229     // qDebug()<<sg1;
230     //check for each 9 cell groups sum
231     sum=0;
232     for(int l=0; l<2; l++)

```

check()

line 201-214 checks for ~~horizontal~~ rows for their sum. If sum is not 45 line 211 if sum is not 45 i.e. not {1,2,3,4,5,6,7,8,9}. reinitialize lbl as "Incorrect" and sg0 as 1. sg0, sg1, sg2 are 1's signals if there are incorrect sums.

```

window.cpp @ P7proj - Qt Creator
File Edit View Build Debug Analyze Tools Window Help
Projects
P7proj
  CMakeLists.txt
  P7proj
    Header Files
      window.h
    Source Files
      main.cpp
      window.cpp
      game.txt
      mainwindow.ui
  CMake Modules
Open Documents
main.cpp
window.cpp
window.h
P7proj
Release

214     }
215     // qDebug()<<sg0;
216     // check for vertical sum
217     sum=0;
218     for(int j=0; j<9; j++)
219     {
220         for(int k=0; k<9; k++)
221         {
222             sum += data[j+(9*k)].toInt();
223         }
224         if(sum!=45) { lbl= "Incorrect";sg1=1;
225             update(); }
226         else{ sum=0; }
227     }
228
229     // qDebug()<<sg1;
230     //check for each 9 cell groups sum
231     sum=0;
232     for(int l=0; l<2; l++)
233     {
234         for(int i=0; i<2; i++)
235         {
236             for(int j=0; j<3; j++)
237             {
238                 for(int k=0; k<3; k++)
239                 {
240                     sum += data[k+(9*j)+(3*i)+(27*l)].toInt();
241                 }
242             }
243             //qDebug()<<sum;
244             if(sum!=45) { lbl= "Incorrect";sg2=1;
245                 update(); }
246             else{ sum=0; }
247         }
248     }
249     qDebug()<<sg2;
250     if(((sg0!=1)&&(sg1!=1)&&(sg2!=1))==true){lbl= "You won!";}
251 }
252 }

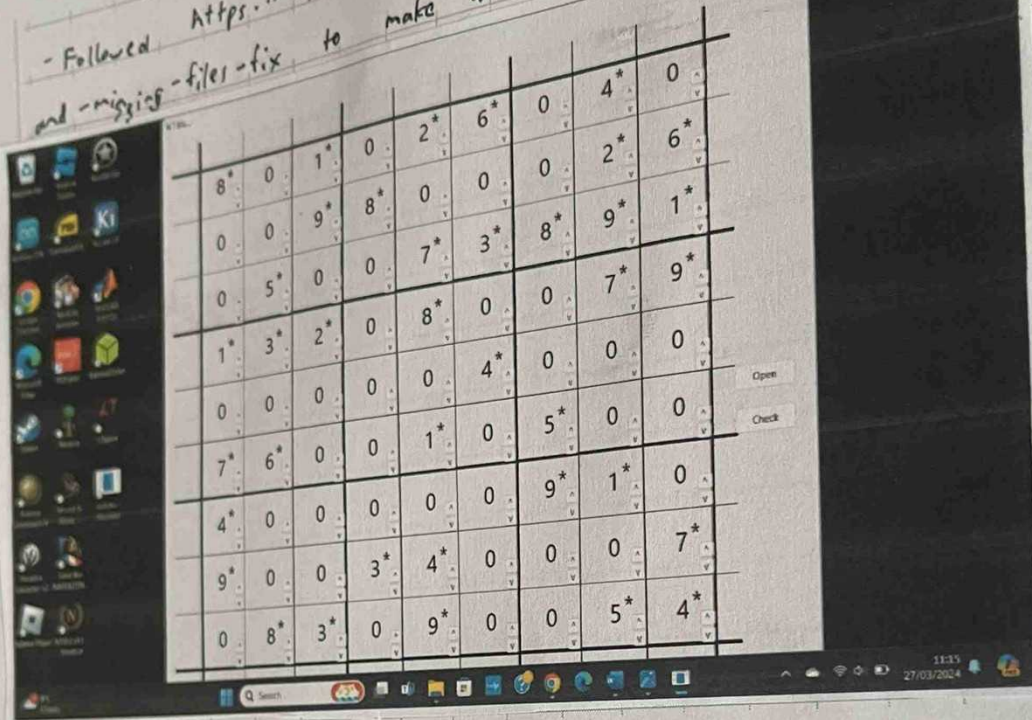
```

217
line 218-227
checks for
columns

line 231-248
checks for
the 9 cell groups

line 251 if
all sg0, sg1, sg2
are 0 then
reassign lbl
to "You won!"

- Followed <https://the-deckie-intj.hashnode.dev/create-a-qt-project-executable-and-missing-files-fix> to make an application of the game.



Final GUI of Sudoku app.

① The measured value is really close

2.3. Measuring Output Impedance
The initial $V(x)$ is 3V.

- Connected a $2k\Omega$ potentiometer.

- Varied the potentiometer until $V(x)$ is 1.5V.

- The output impedance is $2.1 k\Omega$

20/March/24

P7: Sudoku GUI 2

2. Preparation

2.1. Read the Qt tutorial on GUI design
Read the Qt 5 tutorial at <http://zetcode.com/gui/qt5/>

- Read the Qt 5 tutorial at <http://zetcode.com/gui/qt5/>

① A signal is emitted when a particular event occurs
A slot is a normal C++ method; it is called

when a signal connected to it is emitted.

② The upgraded GUI should look like a 9×9 grid, with the buttons, Open and Check, on the right side. The 9×9 Each cell in the grid has an up and down button to change the cell number.

③ Signals are mouse clicks from the user.

Slots are Open(), Plus(), Minus(), & Check().

Open() let lets you choose and open a text file as the initial game data in the relevant directory.

Plus() increases the cell number when the up button is clicked

Minus() decreases the cell number when the down button is clicked

Check() checks if the numbers in the grid match the correct solution.

- Downloaded and unzipped P7 sample code.

- Opened Qt Creator on laptop.

- Created a new project P7proj.

- Copied & pasted, and replaced P7 sample code into P7proj and

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- Replaced the files -
- Copied the code from P6 and modified it in P7 proj -
- 2.2. Consider how to write to a GPIO pin
- Considered how to write to a GPIO pin of the Raspberry Pi
- Installed WiringPi

3. Laboratory Work

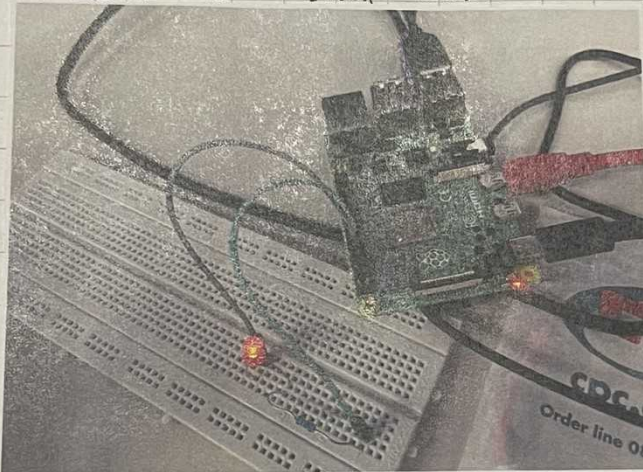
3.1. Write to a GPIO pin

For

- Connected RPi to the laboratory computer.
- Made the connections as shown below.

Lab Code:	None	Poor	OK	Good	Exceptional	Outstanding
<u>Preparation</u>	0	1	2	3	4	5
<u>Progress</u>	0	1	2	3	4	5
<u>Understanding</u>	0	1	2	3	4	5
<u>Logbook Use</u>	0	1	2	3	4	5
Student username:	Hv2n23					
Student Full name:	Timoteo Volante					
Demonstrator only						
Signed:						R.

0.85



WiringPi

- Installed ^{WiringPi} Raspberry gpio / c - wiringpi - setup. | <https://learn.sparkfun.com/tutorials/>

- Entered `cd wiringpi`
- `gpio -g mode 17 output`
- `gpio -g write 17 1`

to ~~for~~ turn on the LED.

- Made the text document `led.cpp` inside `wiringpi` folder.
- Entered `gcc -o led led.cpp -I wiringpi` to compile it.
- Entered `sudo ./led` and the LED blinked.

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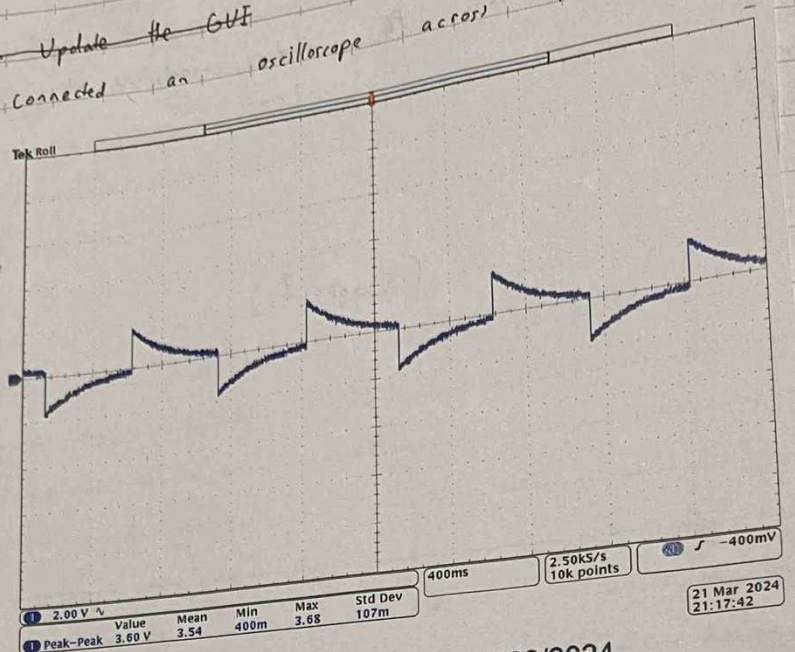
Date

Signed by Witness

Print Name

Date

22. Update the GUI
- Connected an oscilloscope across the LED.



MDO4054B-3 - 15:26:41 21/03/2024

```
# include <wiringPi.h>
```

```
int main()
```

```
{
```

```
wiringPiSetup();
```

```
pinMode(0, OUTPUT);
```

```
for(;;)
```

```
{
```

```
digitalWrite(0, HIGH); delay(500);
```

```
digitalWrite(0, LOW); delay(500);
```

```
}
```

```
return 0;
```

```
}
```

Code to toggle LED using RPi.

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Date

Print Name

Date

3.2. Update the GUI

```

1  #ifndef DATA_H
2  #define DATA_H
3
4  #include <QBrush>
5  #include <QPen>
6  #include <QPixmap>
7  #include <QWidget>
8  #include <QMouseEvent>
9  #include <vector>
10 #include <map>
11
12 #include <QString>
13
14 using namespace std;
15
16 class Data
17 {
18 public:
19     Data();
20     Data(QString DataString);
21     void storedata(QString DataString);
22     QStringList getdata();
23
24 private:
25     QStringList data;
26 };
27
28 #endif // DATA_H
29

```

class data header file

Class Data stores

the game data

```

1  #include "data.h"
2  #include <iostream>
3
4  using namespace std;
5
6  Data::Data()
7  {
8      QStringList DataString;
9      for(int i=0; i<81; i++)
10     {
11         DataString.append("0");
12     }
13     data= DataString;
14 }
15
16 Data::Data(QString DataString)
17 {
18     data= DataString.split(",");
19 }
20
21 void Data::storedata(QString DataString)
22 {
23     data= DataString.split(",");
24 }
25
26 QStringList Data::getdata()
27 {
28     return data;
29 }
30
31

```

class Data cpp file

```

1  #ifndef WINDOW_H
2  #define WINDOW_H
3
4  #include <QBrush>
5  #include <QPen>
6  #include <QPixmap>
7  #include <QWidget>
8  #include <QMouseEvent>
9  #include <QMouseEvent>
10 #include <QPushDialog>
11
12 #include <vector>
13 #include <map>
14
15 #include <QLabel>
16 using namespace std;
17
18 class Window : public QWidget
19 {
20     Q_OBJECT
21 public:
22     Window(QWidget *parent = 0);
23
24 protected:
25     // methods and members
26     void paintEvent(QPaintEvent *event);
27     int l;
28
29 public slots:
30     // event handlers (in Qt "slots")
31     // ...
32     void Open();
33     void Plus();
34     void Minus();
35     void Check();
36
37 };
38

```

widget Window
header file

```

1  #include "window.h"
2  #include "data.h"
3
4  #include <QPainter>
5  #include <QDebug>
6  #include <QTextStream>
7  #include <QMessageBox>
8  #include <QGridLayout>
9  #include <QSpinBox>
10 #include <QPushButton>
11 #include <QFileDialog>
12
13 #include <map>
14 #include <vector>
15
16 #include <QFile>
17 #include <QVector>
18
19 using namespace std;
20
21 Data data1;
22
23 // Main window class
24 Window::Window(QWidget *parent) : QWidget(parent)
25 {
26     // set form size
27     setFixedSize(1000, 1000);
28     setWindowTitle("Bla....");
29
30     // create widgets....
31
32     // open button
33     auto *openBtn = new QPushButton("Open", this);
34     openBtn->setGeometry(900, 450, 75, 30);
35
36     connect(openBtn, &QPushButton::clicked, this, &Window::Open);
37
38     // plus&minus buttons
39

```

widget Window
cpp file

← Creates the
Open button

```

39  QStringList DataList= data1.getdata();
40  l=0;
41  for(int k=0; k<9; k++)
42  {
43      for(int j=0; j<9; j++)
44      {
45          if(DataList[l]=="0"){
46              auto *plsBtn1= new QPushButton("=", this);
47              auto *minBtn1= new QPushButton("-", this);
48
49              plsBtn1->setGeometry(130 + j*90, 90 + k*80, 20, 20);
50              minBtn1->setGeometry(130 + j*90, 110 + k*80, 20, 20);
51
52              plsBtn1->setProperty("order", l);
53              minBtn1->setProperty("order", l);
54
55              connect(plsBtn1, &QPushButton::clicked, this, &Window::Plus);
56              connect(minBtn1, &QPushButton::clicked, this, &Window::Minus);
57          }
58      }
59      l++;
60  }
61  //check button
62  auto *checkBtn = new QPushButton("Check", this);
63  checkBtn->setGeometry(900, 500, 75, 30);
64  connect(checkBtn, &QPushButton::clicked, this, &Window::Check);
65  }
66
67  void Window::Plus() {
68      QPushButton *plsBtn = qobject_cast<QPushButton*>(sender());
69      int ord= plsBtn->property("order").toInt();
70      QStringList DataList= data1.getdata();
71      int val= DataList[ord].toInt();
72      val++;
73      DataList[ord]= QString::number(val);
74  }

```

Creates the up & down buttons

Widget Window

cpp file (continued...)

Creates the
Check button.

up & down buttons
change the value
of the number
in a cell

Plus() slot

increases the cell
number.

Minus() slot

decreases the
cell number.

paintEvent() method

updates the
GUI when an
event is triggered

```

77  QString stringdata= DataList.join(",");
78  data1.storedData(stringdata);
79  update();
80  }
81
82  void Window::Minus() {
83
84      QPushButton *plsBtn = qobject_cast<QPushButton*>(sender());
85      int ord= plsBtn->property("order").toInt();
86      QStringList DataList= data1.getdata();
87      int val= DataList[ord].toInt();
88      val--;
89      DataList[ord]= QString::number(val);
90      QString stringdata= DataList.join(",");
91      data1.storedData(stringdata);
92      update();
93  }
94
95  void Window::paintEvent(QPaintEvent * event)
96  {
97      QPainter painter(this);
98      QPen pen, pen2;
99      QFont font;
100     font.setPixelSize(30);
101     pen.setWidth(1);
102     pen2.setWidth(4);
103     painter.setPen(pen);
104     painter.setFont(font);
105     painter.setPen(pen2);
106     QPoint p3(80, 20), p4(80, 900);
107     painter.drawLine(p3, p4);
108     QPoint p5(20, 60), p6(900, 60);
109     painter.drawLine(p5, p6);
110
111     QStringList DataList= data1.getdata();
112     int i=0;
113     for(int k=0; k<9; k++)
114     {

```

```

133 for(int k=0; k<9; k++)
134 {
135     for(int j=0; j<9; j++)
136     {
137         if((j==2)&&(j==5)&&(j==8)) { painter.setPen(pen2); }
138         if((j==2)&&(j==5)&&(j==8)) { painter.setPen(pen1); }
139         else { painter.setPen(pen1); }
140         QPoint p1(130+j*20, 20), p2(130+j*20, 900);
141         painter.drawLine(p1, p2);
142     }
143     painter.setPen(pen2);
144     painter.drawText(100 + j*20, 100 + k*20, DataList[i]);
145     i++;
146 }
147
148 for(int k=0; k<9; k++)
149 {
150     if((k==2)&&(k==5)&&(k==8)) { painter.setPen(pen2); }
151     else { painter.setPen(pen1); }
152     QPoint p1(20, 130+k*20), p2(900, 130+k*20);
153     painter.drawLine(p1, p2);
154 }
155
156 painter.setPen(pen2);
157 QPoint p1(70, 20), p2(70, 920);
158 painter.drawLine(p1, p2);
159 QPoint p3(20, 770), p4(900, 770);
160 painter.drawLine(p3, p4);
161
162 void Window::Open()
163 {
164     QString fileName = QFileDialog::getOpenFileName(this,
165     tr("Open File"), "C:/Users/theov/OneDrive/Documents/EEE2023-2024/Semester2/Advanced Programming/3rd year/PT/PTproj", tr("Text File"));
166     QFile file(fileName);
167 }

```

```

149 file.open(QIODevice::ReadOnly | QIODevice::Text);
150 QTextStream in(&file);
151 QString DataString;
152 DataString = in.readAll();
153 data1.storeData(DataString);
154 file.close();
155
156 void Window::Check()
157 {
158     QStringList DataList = data1.getData();
159     //check for horizontal sum
160     int sum=0;
161     int k=0;
162     for(int j=0; j<9; j++)
163     {
164         for(int i=0; i<9; i++)
165         {
166             sum += DataList[k+i].toInt();
167         }
168         if(sum!=30)
169         {
170             QPainter painter(this);
171             QFont font;
172             QPen pen;
173             font.setPixelSize(30);
174             pen.setWidth(4);
175             painter.setPen(pen);
176             painter.setFont(font);
177             painter.drawText(900, 400, "You lose");
178             //exit(0);
179         }
180     }
181 }

```

Open() store slot
let you choose and
open a text file
or initial game
data when the
open button is clicked.

Check() slot checks if the numbers in the grid
is the correct solution when the check button is clicked.

```

186 }
187 else{sum=0;}
188 k++;
189 }
190 // check for vertical sum
191 sum=0;
192 for(int j=0; j<9; j++)
193 {
194     for(int i=0; i<9; i++)
195     {
196         sum += DataList[j+(9*i)].toInt();
197     }
198     if(sum!=30)
199     {
200         QPainter painter(this);
201         QFont font;
202         QPen pen;
203         font.setPixelSize(30);
204         pen.setWidth(4);
205         painter.setPen(pen);
206         painter.setFont(font);
207         painter.drawText(900, 400, "You lost.");
208         //exit(0);
209     }
210     else{sum=0;}
211 }
212 //check for 6 cells sum
213 QPainter painter(this);
214 QFont font;
215 QPen pen;
216 font.setPixelSize(30);
217 pen.setWidth(4);
218 painter.setPen(pen);
219 painter.setFont(font);
220 painter.drawText(900, 400, "You won!");
221 update();
222 }

```

ES
C1 0.76 57. 3.33%

0	1	0	0	0	0	0	0	6
7	0	3	0	0	4	0	0	8
0	0	0	6	0	0	3	0	0
0	0	2	8	4	0	0	1	0
0	4	0	1	0	6	0	7	2
0	0	1	5	0	0	6	9	0
0	0	0	3	0	0	4	0	9
0	8	0	0	0	0	0	0	0
3	9	6	0	5	0	0	0	1

Graphical User Interface
of Sudoku

P0
P1
P2 0.95
P3 1
P4 0.875
P5 0.7
P6 0.875
P7 0.85

AP = 26.25%

T7 0.875 3.3%

X10 0.825 1.7%

ES = 14.02%

T8 0.75 3.3%

X4 0.925 1.7%

13.55%

X6 0.975 1.7%

T6 0.9 5%

X7 0.6 1.7%
A12 0.8 6.7%

ES C1 0.76 3.33%
A13 0.7 5%

ES = 22.41%

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